Computers chem. Engng, Vol. 16, No. 4, pp. 293–297, 1992
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0098-1354/92 $5.00 + 0.00
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SHORT NOTE

NEURAL NETWORK FORECASTING OF SHORT, NOISY TIME SERIES

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(Received 23 January 1991; final revision received 1 October 1991; received for publication 10 December 1991)

Abstract—The use of neural networks to forecast short, noisy time series was investigated by comparing neural networks used as function approximators for individual time series with neural networks used to optimally combine traditional forecasting methods based on training across a group of series. Neural network forecast performance on 384 economic and demographic time series was compared to two traditional alternatives: linear regression and the even average of six exponential smoothing methods. On these kinds of partially stochastic series, the use of neural networks as function approximators was found to generally give significantly less accurate forecasts than linear regression. The use of pre-processing to remove seasonal variation improved the forecast accuracy of neural networks, but an even average of exponential smoothing methods was still more accurate. In contrast, the use of neural networks to combine traditional forecasts gave small but significant performance improvements over conventional forecasting methods. Neural network combining of forecasts limits overfitting and, furthermore, can extract general rules of forecasting based on examples across many different time series, making it a potential alternative to rule-based systems.

INTRODUCTION

Modeling and forecasting in chemical engineering involve two broad classes of problems: deterministic problems such as chemical processes where physical models and extensive measurements are often available, and complex stochastic problems such as forecasting inventory levels, competitor pricing or utility loads where definitive physical models and extensive data are lacking. In the former case, techniques of model-based system identification may be appropriate, while in the later case empirical methods such as exponential smoothing or ARMA are often used.

Neural networks have been proposed as an alternative to both types of models, and have already showed promise as models for a wide variety of largely deterministic problems in relatively few variables (Admoaitis *et al.*, 1990; Bhat and McAvoy, 1989, 1990; Lapedes and Farber, 1987). Applications of neural networks for forecasting time series which are less clearly deterministic have, however, not met with clear success—perhaps because the processes studied were purely random or the data and models were in some way inadequate (Reinschmidt, 1990; White, 1989). For problems which depend on very noisy economic processes, e.g. the forecasting of purchase orders, improvements in forecast accuracy may be inherently limited to some few percentage points, but such small improvements are often highly significant economically. For instance, in forecasting utility loads, even at the company level the size of the resources involved may be such that a few percent gain in forecast accuracy can represent thousands of dollars per day.

We present the results of an investigation of using neural networks for forecasting. Our results indicate that the conventional neural network forecasting methods which work well on clearly deterministic problems are not appropriate for problems with few and noisy data. We find it is more effective to use neural networks trained on groups of series to choose the optimal combination of a set of conventional forecasts, with networks learning in effect to predict when different forecasting methods work well. This approach makes use of the small number of parameters in conventional forcasting methods, while incorporating the flexibility of neural networks in a restricted manner.

METHODS

Two neural network forecasting techniques are compared which represent extremes: direct forecasting, which uses a very general network form, and

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network combining, which uses a highly-structured form. Both remain fundamentally similar in that they rely on an optimization process and may involve arbitrary inputs. Below we outline the two techniques and introduce the approach used to evaluate their performance.

Network forecasts

Direct forecasting is the method used successfully in the literature to forecast chemical process data by developing an empirical process model. In this method feedforward backpropagation neural networks (Rumelhart and McClelland, 1987) are trained to represent a process as a time delay mapping:

$$\tilde{x}(t) = F[x(t-\Delta), x(t-2\Delta) \ldots x(t-n\Delta), \omega], \quad (1)$$

where $\tilde{x}(t)$ is the approximation at time t of $x(t)$ which is a vector or scalar time series of process data and ω is a vector of model parameter or weights which are chosen to minimize an error:

$$\min_{\omega} \sum_{t'<t} [x(t') - \tilde{x}(t')]^2. \quad (2)$$

The sum is over the historical data or possibly over the historical data after some smoothing or other pre-processing. Forecasts more than Δ time into the future are made by iteration of the equation. See Farmer and Sidorowich (1988) regarding the merits of iteration vs minimization for arbitrary times into the future. The objective function may be minimized by any number of techniques, but we chose conjugate gradient minimization. Note that with this method the network form is quite general and is in theory capable of approximating any function to arbitrary accuracy.

A second, more structured method, which we call network combining, uses neural networks to create weighted averages of conventional forecasts. The conventional forecasts are computed by standard algorithms such as exponential smoothing and involve relatively few free parameters. Each conventional forecast represents a different implicit assumption about time series, e.g. time series are basically linear, exponentially discounted or regress to a mean. The network outputs for network combining are constrained to add to one, so a choice is made between the methods. The basis of an optimal choice is forecast accuracy across a group of training series. The network is thus used as a selector of methods rather than as a function approximator.

The network forecast is:

$$\tilde{x}(t, \omega) = \sum_{i=1}^{n} \frac{y_i(\omega)}{\sum_{j=1}^{n} y_j(\omega)} \hat{x}_i(t), \quad (3)$$

where $\tilde{x}(t, \omega)$ is the approximation of $x(t)$ with network weights ω, $y_i(\omega)$ is the ith output of the network and $\hat{x}_i(t)$ is the forecast of conventional method i. Note that the network outputs may be a function of inputs which are features of a time series, such as its deviation about a mean.

The minimization problem is:

$$\min_{\omega} \sum_{\text{training series}} [x(t) - \tilde{x}(t, \omega)]^2, \quad (4)$$

where t is the time into the future over which forecasts are being made. Figure 1 shows diagrams of both network methods.

The two methods differ in several important respects. Direct forecasting uses a very general network form to learn a model of a single process, while network combining uses precomputed features of a series to learn how to accurately forecast a set of time series by combining conventional forecasts. Network combining uses a more structured network making solutions easier to interpret and requiring fewer examples for training (Mavrovouniotis and Chang, 1992; Psichogios and Ungar, 1991). Also, any desirable properties of the precomputed features such as insensitivity to noise are potentially conferred upon the network. Finally, the use of the structured network creates a bridge between rule-based forecasting techniques (Collopy and Armstrong, 1990; Rahman, 1990) and neural network ones. The same features

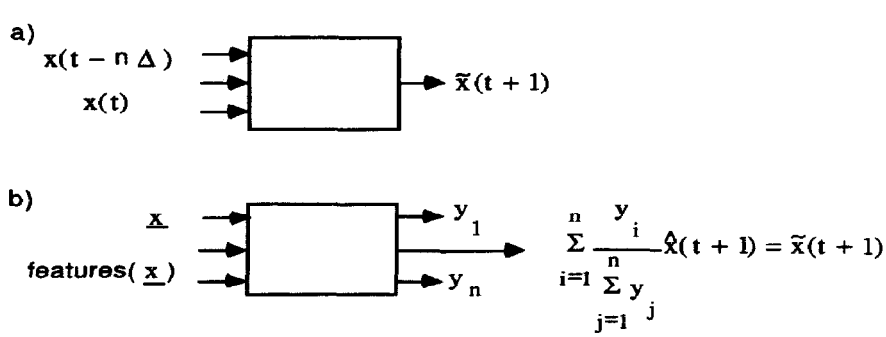

Fig. 1. (a) Direct forecasting. (b) Network combining architectures.

used by forecasting rules may simply be fed in as inputs to a neural network or operated on by network outputs. The advantage of the network is that an automated optimization rather than a human expert determines the action to be taken with the feature information.

Performance evaluation

The measures of forecast performance are based on forecast accuracy across groups of series since good performance on a single series is not a strong statement regarding a method. Evaluation of the different methods was done using the yearly and quarterly subsets of a publically available and extremely well-studied collection of economic and demographic time series (Armstrong and Lusk, 1983; Makridakis *et al.*, 1982). These series come from very diverse sources and have different starting and ending dates and sampling times. The yearly sampled subset contained 181 time series; the quarterly 203 time series. With respect to forecasting, similarities are expected to exist between these series and series taken from chemical engineering applications such as loads, but we were unable to find a sufficiently large, diverse and widely-accepted data set in that area. In this study the median average percentage error measure (MdAPE) (Makridakis *et al.*, 1982) and the geometric mean relative absolute error (GMRAE) (Armstrong and Collopy, 1991) were used. The MdAPE represents a mean percentage forecast error while the GMRAE compares forecast performance with that of the no-change forecast—values less than one denote performance better on average than the no-change forecast. These measures are considered to be more reliable than squared error measures (Armstrong and Collopy, 1991).

To fairly assess the performance of neural network methods they must be trained and tested on different data. Half a group of time series were repeatedly pseudorandomly chosen for training, as were the network initial weights. For direct network forecasting, networks were trained on the historical data of a series [all but the last six points for annual series and all but the last eight points for quarterly series, following Makridakis *et al.* (1982)] and then tested on prediction data from the same series. For network combining, the network was trained for each prediction horizon separately using points from the training group as a series and then was tested on the remaining half of the series. The MdAPE and GMRAE measures were computed on forecasts of unused data and the process repeated 100 times for each network method. Means of the MdAPE and GMRAE measures across the trials were done and comparisons of the means of different methods using the t-test were performed at the 1% level of significance. This leads to a highly reproducible assessment of neural network forecast performance.

RESULTS

The results presented here compare the performance of different network forecasts with traditional methods. Variables include the network architecture, the use of preprocessing, and the use of structured vs unstructured networks. The conventional forecasts for comparison are the least-squares line and a method called combining A which is a straight average of six different exponential smoothing techniques and which performed well in previous comparisons (Makridakis *et al.*, 1982).

Table 1 shows direct neural network forecast performance for various architectures. An architecture of 4, 3, 2 denotes a four-input node three hidden and two output node network. One of the input nodes is a bias. For the error measures listed, lower values denote better forecasts. All direct network forecasts are significantly worse than combining A, and most are worse than linear regression. It is clear that increasing the number of hidden nodes improves forecast performance even when the number of weights exceeds the number of data points used in training. (The yearly series have an average of 19 points available for training; the 5, 10, 1 network had

Table 1. Direct neural net forecasting on 181 yearly series

	Mean MdAPE	Mean GMRAE
Linear regression	15.5	0.88
Combining A[a]	12.0	0.71
Networks: (5, 1 denotes 5 inputs and 1 output, etc.)		
5, 1	17.4	0.98
5, 3, 1	16.0	0.98[b]
5, 10, 1	14.9[c]	0.83[c]
8, 1	17.9	1.04
8, 3, 1	17.4	0.99
8, 10, 1	16.2	0.92

[a] Better than all alternatives at 1%.
[b] Not different from linear regression at 1%.
[c] Better than linear regression at 1%.

Table 2. Direct neural nets—effects of preprocessing on 203 quarterly series

	Mean MdAPE	Mean GMRAE
Linear regression	10.6	0.86
Combining A[a]	9.5	0.85
Networks: (5, 1 denotes 5 inputs and 1 output, etc.)		
No preprocessing:		
5, 1	14.0[d]	1.15[d]
5, 3, 1	11.0[d]	0.95[d]
5, 10, 1	10.7[c]	0.90[c]
Deseasonalized prior to training:		
5, 1	11.6[d]	0.95[d]
5, 3, 1	10.2[b]	0.93[b]
5, 10, 1	10.2[c]	0.91[c]

[a]Better than all alternatives at 1%.
[b]Not different from linear regression at 1%.
[c]Better than linear regression at 1%.
[d]Worse than linear regression at 1%.

61 weights.) By contrast an increase in the number of input nodes from 5 to 8 decreased forecast accuracy for all architectures tested. These results show the utility of hidden nodes for finding relevant features of the time series and the danger of input nodes as contributors to overfitting.

The use of hidden nodes can be further explored by comparing the performance of networks trained on preprocessed data to those trained on raw data. Table 2 shows that when networks were trained on quarterly time series which had been deseasonalized, the gains in performance with an increase in the number of hidden nodes were less. This illustrates how components of a time series can be captured by either a network or a fixed algorithm such as deseasonalizing.

Finally, Table 3 shows that network combining can significantly outperform linear regression and combining A. In the case shown a 1-input 6-output net was used to produce a weighted average of the same size exponential smoothing methods whose even average is combining A. With a constant bias as input, the network combining is based only on the optimal weights found during training. For these sets of time series no feature inputs were found which significantly improved performance beyond the bias only case. It is, however, easy to show for other sets of time series that features are useful in improving network models.

The results of network combining are intuitively appealing. For instance, the better performance of network combining over combining A was accompanied by the zeroing of the contributions of the two exponential smoothing methods which appeared to be the worst of the six. Also, in combined forecasts of the no-change or random walk forecast and Holt's exponential smoothing method, the network placed increasing weight on the random walk forecast as the forecast horizon extended. Furthermore, in going from yearly, to quarterly, to monthly time series—increasing weight was again placed on the random walk forecast. This suggests that very much like a forecasting expert, the network combining learned to rely more heavily on the conservative no-change forecast as the forecast extended farther into a future or was applied to noisier data.

CONCLUSIONS

The generality of neural networks makes them attractive for creating models of a wide variety of processes from example data. This generality, however, can lead to overfitting. For the complex and noisy processes studied here this seems to be the case. In this study, the more general neural network forms used in direct forecasting did worse than traditional forecasting methods having relatively few adjustable parameters. On the other hand, neural networks with the structured flexibility used in network combining did significantly better than the less flexible traditional methods to which they were compared. For those problems in chemical engineering which are deterministic and for which extensive data are available, the use of neural networks as function approximators is likely to be effective. However, for problems involving complex interactions where adequate data are not available, a structured network approach emphasising preprocessing to remove

Table 3. Network combining—forecasts on 181 yearly series

	Mean MdAPE	Mean GMRAE
Linear regression	15.5	0.88
Combining A	12.0	0.71
Network: (1, 6 denotes 1 input and 6 outputs)		
1, 6	10.0[a]	0.60[a]

[a]Better than all alternatives at 1%.

known functionalities and the use of noise-insensitive features may be more appropriate. Problems with economic or weather-related inputs are likely candidates.

On the data investigated here, network combining was seen to capture some of what might be seen as forecasting expertise. With a simple network structure and the choice of some relevant features to operate on, neural network mappings become much more like rules. With neural networks, however, the "rules" may be automatically induced from data rather than being elicited from experts and then coded. Furthermore, a neural network does "rule" development and validation simultaneously and is as impartial as the optimization process allows. These capabilities may make structured neural networks an attractive alternative to rule-based systems.

The optimal amount of data to provide a neural network and the optimal structure to build into it depend on the particular application. For the noisy data considered here only individual time series were available, other correlated series would probably not have improved predictions, and a network constrained to give a weighted combination of conventional forecasts worked well. In contrast, in forecasting, for example, electric utility loads the time series of dry and wet bulb temperatures, relative humidity, wind speed and direction, and cloud cover may all be relevant (Rahman, 1990) and for this more complex problem the constraints to apply are less clear. Finally, in forecasting behavior of chemical process units, time series from many different units and controllers can be useful, and constraints such as mass and energy conservation can be added to improve learning speed and modeling accuracy (Psichogios and Ungar, 1991). Interestingly, as the degree of structure used in neural networks increases, the distinctions between the neural networks, rule bases and constrained optimization begin to blur. With increasingly cheap data handling and computation available, methods of automatically creating reliable models from data will become increasingly attractive economically. Used cautiously and with care given to performance evaluation, neural networks remain a promising candidate for these applications.

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